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Sequential extraction of *Hericium erinaceus* using green solvents

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ABSTRACT

Hericium erinaceus was processed to obtain soluble fractions using a sequence of stages consisting on: microwave hydrogravity (400 W), supercritical CO_2 extraction (20 MPa, 40 °C, 10% ethanol) and the remaining raffinates were subjected either to enzyme assisted extraction (3% protease + cellulase + pectinase, 50 °C, 3 h) or to non-isothermal autohydrolysis with water (200 °C). More than 40% of the mushroom could be solubilized with the proposed green processes, which provided extracts containing the phenolic and the polysaccharide fractions. The phenolic components could be further concentrated by adsorption onto polymeric resins and desorption with 96% ethanol to yield concentrates with ABTS radical scavenging capacity equivalent to 0.6 g of Trolox/g extract.

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1. Introduction

Hericium erinaceus, known as Lion's Mane Mushroom or Hedgehog Mushroom, is commonly used for medicine or food purposes and has attracted recent attention due to its health effects. Ethanol extracts have shown DPPH radical scavenging activities and extended the induction period of lipid oxidation in corn oil emulsion, in lard (Fui, Shieh, & Ho, 2002) and in linoleic acid emulsion (Mau, Lin, & Song, 2002). Methanol extracts (Wong, Sabaratnam, Abdullah, Kuppusamy, & Naidu, 2009) and its solvent fractions exhibited ferric reducing/antioxidant power and radical scavenging activity against $ABTS^{\cdot+}$, $DPPH^{\cdot}$, and nitrite (Li et al., 2012). Boiling water extracts showed relatively high antioxidant index, on the basis of DPPH radical scavenging activity, β-carotene bleaching, inhibition of lipid peroxidation, reducing power and *in vitro* angiotensin converting enzyme inhibitor (Abdullah, Ismail, Aminudin, Shuib, & Lau, 2012) and protected from ulcer in rats by preventing the depletion of antioxidant enzymes (Wong et al., 2013). Various extracts of this mushroom inhibited the growth of pathogenic bacteria (Wong et al., 2009), chloroform fractions of the ethanolic extract were anti-inflammatory (Kim, Lee, Oh, & Rhee, 2012) and the methanol extract showed hypoglycemic effects (Wang, Hu, Wang, Chen, & Chia, 2005). Hot water extracts and ethanolic extracts reduced colon tumor weights in mice (Kim, Kang, Kim, Nam, & Friedman, 2011). Some active novel compounds isolated from methanolic extracts, i.e. hericerin and hericenones, protected against neurodegenerative diseases (Mori et al., 2008) and presented antithrombotic effects (Mori et al., 2010).

The sequential extraction of mushroom antioxidants and antimicrobial agents using solvent of increasing polarity was proposed (Smolskaitė, Venskutonis, & Talou, 2015). Microwaves aided in the extraction of phenols from wild edible mushrooms (Kim et al., 2011; Özyürek, Bener, Guclu, & Apak, 2014). The microwave-assisted extraction (MAE) process is suited for operation with different solvents. However, the use of organic toxic solvents should be limited for food purposes. A simple technology is based on microwave irradiation and earth gravity (microwave hydrodiffusion and gravity, MHG), and consists on the diffusion of the extract outside the plant material and its downward collection. This mode of operation is suited for samples with high water content and for the extraction of volatile compounds (Bousbia, Vian, Ferhat, Meklati, & Chemat, 2009; Flórez, Conde, & Domínguez, 2015). Combination of microwave assisted water extraction and other technologies can be advantageous (Flórez et al., 2015; Pérez, Conde, & Domínguez, 2014) and comply with green chemistry and sustainability criteria (Barba, Grimi, & Vorobiev, 2014; Deng et al.,

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2015). Supercritical CO_2 extraction and subcritical water extraction or autohydrolysis are green technologies proposed for the solubilization of phenolic fractions. Hydrothermal processing has been successfully used for some mushrooms (Matsunaga et al., 2013; Seo & Lee, 2010), but no information on *Hericium* species is available.

Most of the therapeutic polysaccharides found in *H. erinaceus*, are β-glucans (Han, Ye, & Wang, 2013; Khan, Tania, Liu, & Rahman, 2013; Zhu et al., 2014), but mushroom also contain other bioactive compounds, i.e. phenolics, tocopherols, ascorbic acid and carotenoids. The previous separation of other components enhanced the purification of saccharidic compounds (Vegas, Alonso, Domínguez, & Parajó, 2005), providing an additional stream with valuable compounds that could benefit the economy of the process.

Limitation of organic toxic solvents and the integral fractionation of raw materials are desirable. The objective of the present work is to evaluate the feasibility of a sequence of stages with green solvents (including water extraction, microwave hydrogravity, and supercritical carbon dioxide extraction) to fractionate *H. erinaceus*. Further aqueous extraction either aided by hydrolytic enzymes or under subcritical conditions were also tried for the solubilization of the phenolic and saccharidic components from the remaining solid.

2. Materials and methods

2.1. Material

H. erinaceus samples were cultivated by Hifas da Terra (Pontevedra, Spain). Strains were cultured on potato dextrose agar medium (PDA: potato 200 mg/mL glucose 20 mg/mL and agar 20 mg/mL) and incubated at 25 °C for 7–10 days (Fen et al., 2014). These cultures were used to seed *Populus alba* autoclaved wood pellets previously humidified. Mycelium was allowed to grow on wood pellets for 7–10 days at 25 °C, and fruit bodies developed after 20 days. Fresh fruit bodies were cut into pieces of approx. 2 cm × 2 cm before use. The average moisture content of the fresh mushroom was 70.22 ± 1.36% of the wet weight. In dry basis the mushroom contained 7.06% ash, 35% carbohydrates, 26% proteins and 28% ethanol extractives.

Sepabeads SP700 was purchased from Resindion S.R.L. (Mitsubishi Chemical Corp.).

2.2. Extraction and processing technologies

2.2.1. Microwave hydrogravity extraction (MHG)

The fresh mushrooms were contacted with distilled water overnight at 4 °C. The solid and liquid fractions were separated by filtration through filter paper and the solids were then processed by microwave hydrogravity extraction (MHG) at irradiation powers in the range 70–900 W. The drained liquid was collected and analyzed for total solubles, total phenolics and ABTS radical cation scavenging activity. Extractions were performed at least three times, and the mean values were reported.

2.2.2. Supercritical fluid extraction (SFE)

The freeze-dried mushroom and the MHG processed mushroom were ground in an IKA grinder (M20, Germany) and samples of 10 g were packed with glass beads into a Thar Designs extractor, model SFE-1000F-2-C10 (Pittsburgh, USA) with a 1 L cylinder extractor and two 500 mL separators. The CO_2 was precooled with a water bath (PolyScience model 9506, USA) and pumped with a piston pump (P-200A, Thar Design Inc.). The solvent flowed at 25 g CO_2/min and the cosolvent was pumped by an HPLC pump (Series III, Scientific Systems Inc., USA). The modifier was added in the concentration range 0.5–15% (wt basis). The extraction vessel was covered with polypropylene wool to avoid solid particles from entering the extraction line. Pressure was fixed at 20 MPa and temperature at 40 °C, based on literature (Kitzberger, Smania, Pedrosa, & Ferreira, 2007; Mazzutti et al., 2012). The extract was collected during 2 h in the first separator at ambient temperature and further recovered with 96% ethanol. Ethanol was removed in rotavapor and the extracts were kept under N_2, at −20 °C in the dark until analysis.

2.2.3. Enzyme aided extraction (EAE)

Incubation with a mixture of pectinase:cellulase:protease was performed in triplicate independent assays. Ten grams of fresh mushroom or the solid residue from MHG were contacted with sodium acetate-acetic acid buffer pH 4.5 at liquid to solid ratio 25 (v:w), in 500 mL Erlenmeyer flasks placed in an orbital shaker (INNOVA 4000, New Brunswick) at 50 °C during 3 h. The mixture of commercial enzymes Viscozyme:Celluclast:Flavourzyme (2:1.5:1.5) (Novozymes) was added at an enzyme to substrate ratio of 5% (wt). The suspension was heated at 100 °C for 10 min to inactivate the enzymes and centrifuged (4500 rpm, 8 min) to separate the solid residue. The supernatant was filtered through 0.45 μm membranes and the filtrate was stored at −20 °C in the absence of light until analysis.

2.2.4. Autohydrolysis (NIAH)

The ground wet mushroom or the exhausted solid after MHG were contacted with water (liquid:solid ratio 12:1, w:w) and heated in a 0.6 L Parr reactor (Parr Instr. Co., USA) under non-isothermal autohydrolysis, operating at temperatures up to 200 °C (based on the results of previous unpublished data). The vessel was then cooled and opened and the liquid and solid phases were separated by filtration. The liquid phase was further characterized.

2.2.5. Adsorption–desorption (AD)

The aqueous extract obtained after overnight maceration of fresh *H. erinaceus* with distilled water was contacted with SP700 resins at a resin:liquid ratio 1:25 (w:v). The suspensions were placed in Erlenmeyer flasks at 175 rpm in an orbital shaker at 25 °C for 60 min. After solid liquid separation, the aqueous phase was analyzed and the resin was washed with distilled water to remove the unadsorbed compounds, using a 1:3 moist resin:distilled water ratio (w/v) in an orbital shaker at 175 rpm and 25 °C during 20 min. After solid liquid separation, desorption was performed by contacting the saturated resins at a moist resin:96% ethanol ratio 1:3 (w:v) at 175 rpm and 25 °C during 60 min.

2.3. Analytical methods

The total phenolic content was determined by the Folin–Ciocalteu assay (Singleton & Rossi, 1965) and expressed as Gallic Acid Equivalents (GAE). Glucuronic acid, glucose, arabinose, and the sum of xylose + galactose + mannose were determined by HPLC-RI (1100 series Hewlett–Packard) using a 300 × 7.8 mm Aminex HPX-87H column (BioRad, Hercules, CA) at 60 °C with 0.003 M H_2SO_4 as mobile phase. The concentration of oligosaccharides was determined after acid hydrolysis (4% sulfuric acid, 121 °C, 20 min). Xylose, mannose and galactose were determined with a 300 × 7.8 mm CARBOsep CHO 682 column (Transgenomic, Glasgow, UK) operating at 80 °C using distilled water as the mobile phase.

Mineral content (Ca, Mg, Mn, Zn, Cu, Cr, Pb, Cd) of samples was determined by atomic absorption and by atomic emission (Na, K) spectrophotometry (220 Fast Sequential Spectrophotometer, Varian, Palo Alto, CA) after a previous digestion of 0.3 g of ashes with 10 mL of HNO_3 and 1 mL of H_2O_2 in a microwave station

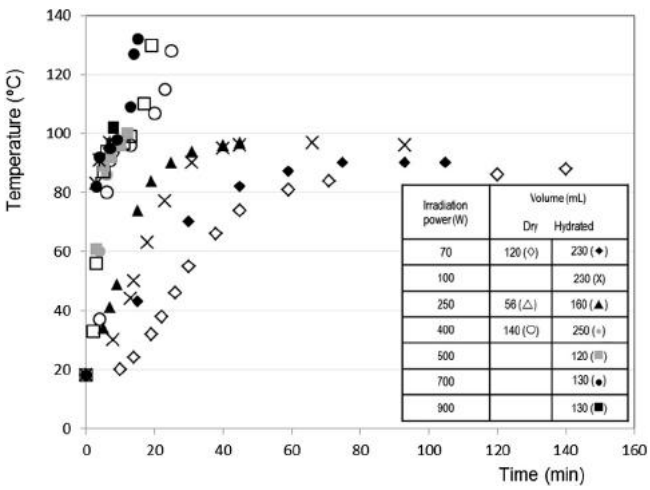

Fig. 1. Effect of irradiation power during MHG on the temperature profile and liquid volume drained for fresh and water-extracted *Hericium erinaceus* samples.

(Marsxpress, CEM), operating at 1600 W during 15 min and maintained 10 min at 200 °C.

Fatty acid methyl esters (FAME) were prepared by trans-methylation (Carreau & Dubacq, 1978). FAMEs were extracted by hexane and, after evaporating the organic phase under reduced pressure, were analyzed in a GC–MS QP 2010 (Shimadzu, Japan) using an SP-2380 column. The temperature gradient increased from 50 °C (2 min) at 40 °C/min to 250 °C, which was maintained for 15 min. FAMEs were identified by comparing their mass spectra with those of authentic standards and NIST MS Search 2.0 library.

The FRAP assay (Benzie & Strain, 1996) was used to measure the ability to reduce the ferric 2,4,6-tripyridyl-s-triazine (TPTZ) complex under acidic conditions. The reagent was made by mixing 25 mL of 300 mmol/L acetate buffer (pH 3.6) and 2.5 mL of a 10 mmol TPTZ/L solution in 40 mmol/L HCl and 20 mmol/L $FeCl_3 \cdot 6H_2O$ in distilled water. Samples (100 μL) were mixed with 3 mL of the reagent, and the absorbance was monitored at 593 nm, using ascorbic acid as standard.

The TEAC assay improved by Re et al. (1999), based on the scavenging of ABTS radical (2,2′-azinobis-(3-ethyl-benzothiazoline-6-sulfonate)) was used. $ABTS^{\bullet+}$ was produced by reacting 7 mM ABTS stock solution with 2.45 mM potassium persulfate, diluted with phosphate buffer saline (PBS) and equilibrated at 30 °C. After addition of 1.0 mL of diluted $ABTS^{\bullet+}$ solution to 10 μL of antioxidant compounds or Trolox standards in ethanol or PBS, the absorbance was read for 6 min. The percentage of absorbance inhibition at 734 nm was expressed as Trolox equivalents.

Mushroom samples were freeze-dried, fixed onto aluminum stubs, sputter coated with gold in an Emitech K550X equipment and examined in a FEI Quanta 200 scanning electron microscope at an accelerating voltage of 12.5 kV.

3. Results and discussion

Carbohydrates are the most abundant fraction in *H. erinaceus* (Zhu et al., 2014), but the water and alcoholic extractives have been studied for their radical scavenging and biological properties and the phenolic and flavonoid components may be the major contributors to the inhibitory effect on reactive species (Mau et al., 2002; Özyürek et al., 2014). Multi-stage extraction methodology involving efficient technologies and green solvents was used to valorizate carbohydrates from mushrooms and to facilitate the purification of the saccharide fractions.

3.1. Microwave hydrogravity (MHG)

The microwave hydrogravity diffusion (MHG) process, based on the application of microwave technology in an open system, was applied to the fresh mushroom biomass (70.22% moisture). Data in

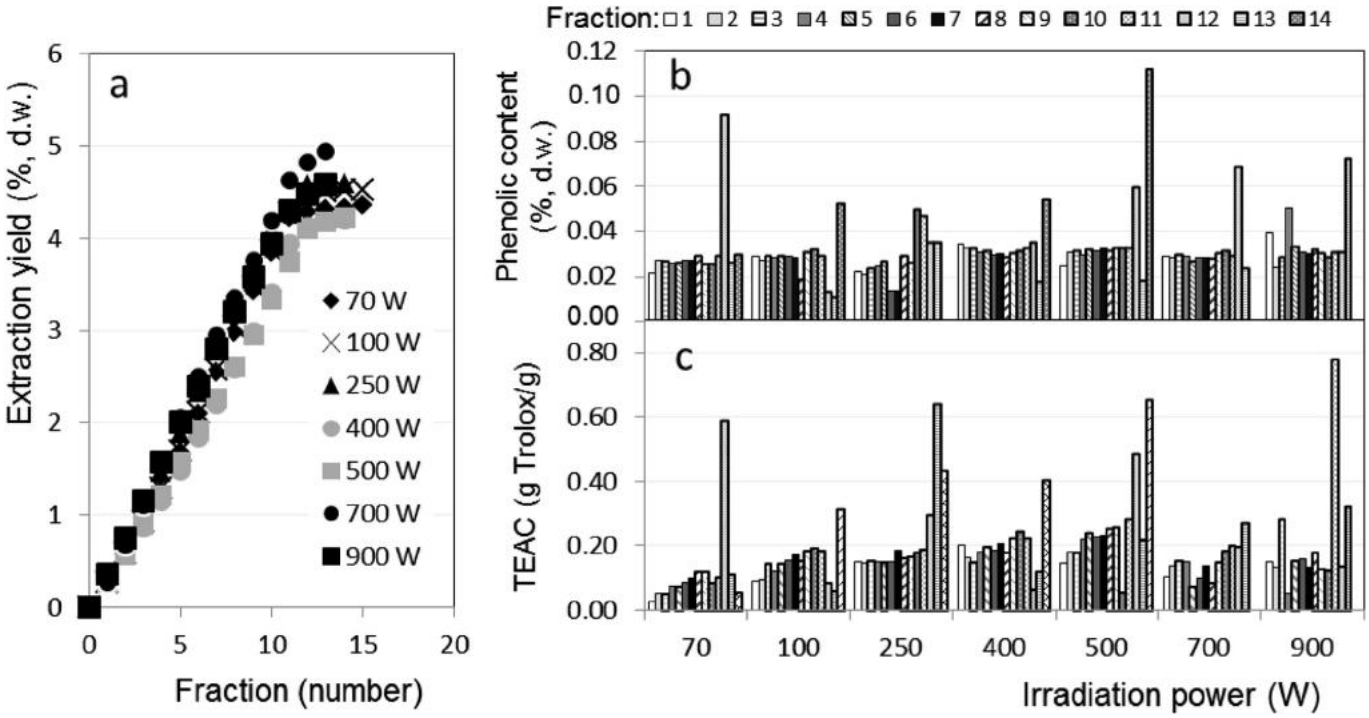

Fig. 2. Effect of irradiation power during MHG on a) extraction yield, b) phenolic concentration and c) TEAC of the fractions from water extracted *Hericium erinaceus*.

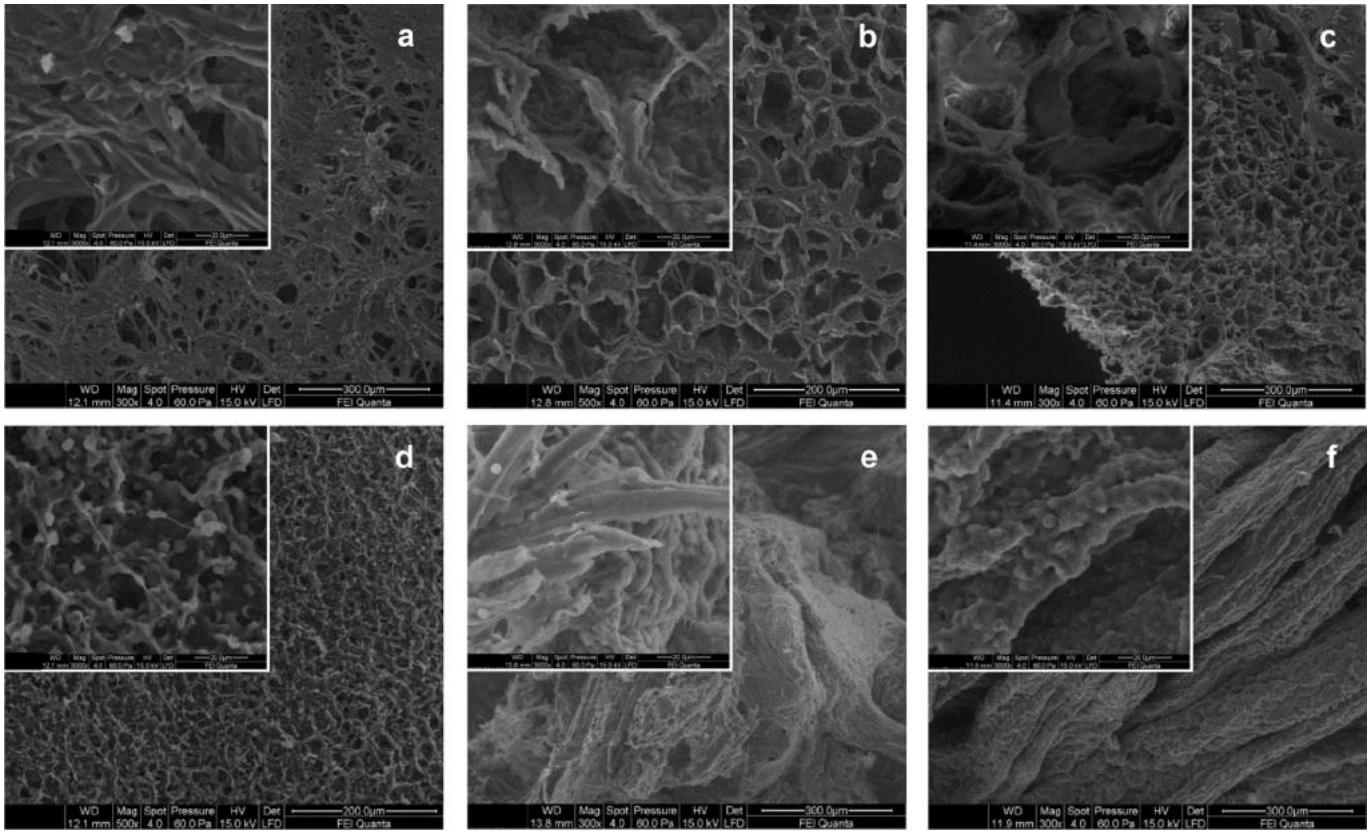

Fig. 3. Microscopic structure of *Hericium erinaceus* internal parts for: a) freeze-dried, b) after MHG at 100 W and c) at 500 W and external parts, d) freeze-dried, e) after MHG at 100 W and f) at 500 W.

Fig. 1 shows the typical temperature profile during MHG, with an initial rapid increase and a plateau until the complete draining of the water phase; further heating beyond this point could led to burning (observed at the higher irradiation powers tested) and was avoided. A previous hydration of mushroom using distilled water (wet mushroom:distilled water ratio 1:2) at 4 °C overnight led to higher yields of drained liquid phase in the MHG due to the higher moisture content and heating efficiency. This behavior was confirmed for irradiation powers in the range studied (Fig. 1). For further studies H. erinaceus was initially hydrated in the selected conditions, after filtration the solids were used for MHG processing. In addition, the separation of the liquid phase allowed the extraction of 5.15 ± 0.69% solubles, containing 1.96 ± 0.11 g GAE (gallic acid equivalents)/100 g extracts with an ABTS radical scavenging capacity equivalent to 0.04 g Trolox/g extract. Up to 42.17 ± 1.49% of these phenolic compounds could be adsorbed onto SP700. During elution with 96% ethanol, 96.72 ± 2.35% of the adsorbed compounds could be efficiently desorbed. The dried desorbed fraction contained 13.30 ± 1.05% phenolics and was equivalent to 0.59 ± 0.05 g Trolox/g.

Fig. 2 shows the extraction yield, phenolic content and antiradical properties of the different fractions extracted for irradiation power in the range 70–900 W, the phenolic content expressed as gallic acid equivalents and ABTS radical scavenging capacity of the fractions collected. The irradiation power has hardly any effect on the extraction yields and on the phenolic content of each collected fraction, except for the higher values found in the latest fractions regardless the power. Slightly higher activity was also noticed on the last fractions and in the extracts from 400 to 500 W. In the short periods needed in this process no decrease associated to thermal

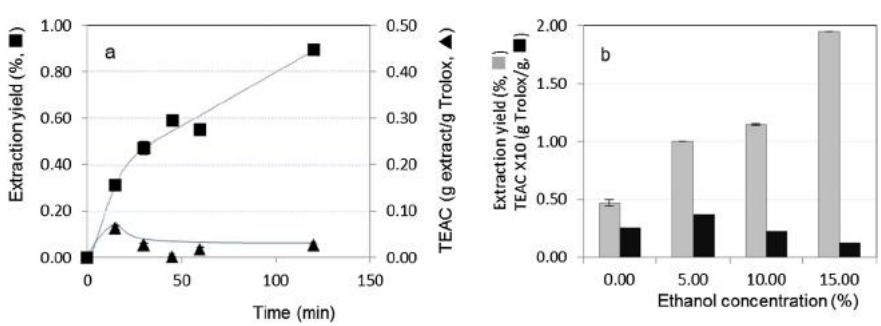

Fig. 4. Supercritical extraction of total solubles and TEAC values of the extracts from freeze-dried *H. erinaceus*: a) extraction kinetics at 20 MPa and 40 °C with pure CO_2 and b) effect of ethanol concentration during supercritical extraction at 20 MPa, 40 °C during 30 min.

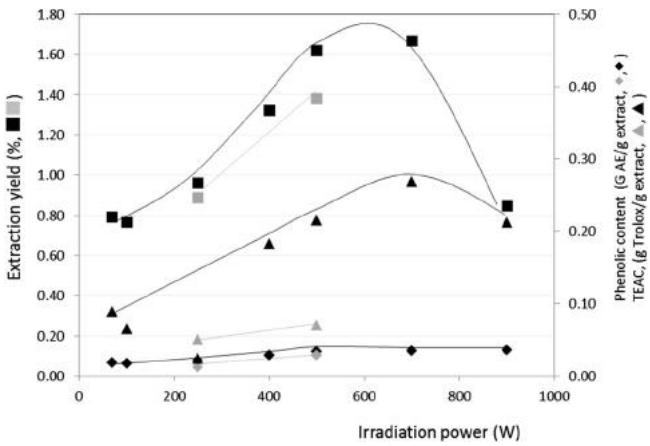

Fig. 5. Effect of MHG irradiation power of water-extracted *H. erinaceus* samples on the extraction yield (squares), phenolic content of the extracts (diamond) and TEAC values (triangle) during supercritical extraction with pure (gray symbol) or with ethanol modified CO_2 (black symbol) of the solids remaining after MHG.

degradation was noticed. During microwave assisted extraction of edible mushrooms with methanol, the antioxidant properties of the extract increased with temperature up to 80 °C, and then decreased, probably due to partial thermal degradation of heat sensitive constituents (Özyürek et al., 2014). The extraction yield was lower than that attained with conventional solvents, reaching 28% with ethanol from this mushroom or with methanol from other mushrooms (25.5–33.1%) (Mau et al., 2002; Wang et al., 2005). The MHG process allowed total phenolics concentrations of 2–3%, comparable to those attained with conventional solvents, i.e. methanolic extracts (12.05–31.20 mg GAE/g extract) (Mau et al., 2002; Wong et al., 2009), chloroform, n-hexane, and n-butanol subfractions from methanolic extracts (11.2–35.2 mg ferulic acid equivalents/g) (Li et al., 2012) or hot water extracts (10.2 mg GAE/g) (Abdullah et al., 2012). The TEAC activity was comparable or higher to that reported for the chloroform subfraction of a methanolic extract from this mushroom, with a phenolic content of 1.1–3.5% being the identified phenolic acids chlorogenic acid, p-hydroxybezoic acid, vanillic acid, ferulic acid, sinapic acid, syringic acid and p-anisic acid and flavonoids (catechin, epicatechin, rutin, myricetin, hesperidin and quercetin) (Li et al., 2012). Benzoic acids, hydroxycinnamic acids and flavonoids have been identified in the microwave-assisted extraction methanolic extraction of polyphenols from edible mushrooms (Özyürek et al., 2014).

The antioxidant capacity of *H. erinaceus* was relatively high among methanolic extracts of several specialty mushrooms (Mau et al., 2002) and among water extracts of several culinary mushrooms (Abdullah et al., 2012). However, the IC_{50} for DPPH was significantly higher for standard antioxidants (Abdullah et al., 2012; Mau et al., 2002), and the inhibition of the lipid peroxidation of phospholipids in egg yolk induced by Fe^{2+} was lower than for quercetin, BHA and ascorbic acid (Abdullah et al., 2012).

The FAME composition of the MHG extracts showed that the most abundant compound was oleic acid (C18:1), accounting for 73.3 ± 0.23% of the total FAME, followed by linoleic acid (C18:2), with 12.0 ± 0.11%, palmitic acid (C16:0), 7.9 ± 0.31% and stearic acid, (C18:0), 6.8 ± 0.03%.

After MHG processing the remaining solid sample was dry (average moisture in the range 10.28% and 8.30%, for samples processed at 100 W and 500 W, respectively) and presented suitable characteristics for stable storage. In addition, the microwave irradiation during these short periods caused structural damage of the cell walls and thus, it could be a suited pretreatment before extraction with apolar solvents. Fig. 3 shows the demolition effect on the cell walls from both internal and external parts of the mushroom exposed to 100 and to 500 W in comparison to a freeze-dried sample.

3.2. Supercritical CO_2 extraction (SFE)

Preliminary experiments with freeze-dried *H. erinaceus* samples (moisture content 8.08%) were carried out to obtain information on the extraction kinetics with supercritical CO_2 at 20 MPa and 45 °C. These operational conditions were selected on the basis of preliminary assays and literature information with other edible mushrooms (Kitzberger et al., 2007; Mazzutti et al., 2012). Due to the technical difficulties in analyzing the phenolic content in those samples, only the radical scavenging properties are shown in Fig. 4. Literature reports on the beneficial effect of the phenolic fraction and for this reason the effect of a polar modifier was assessed during extraction for 30 min under the same pressure and temperature conditions. The use of ethanol increased by almost 4 times the extraction yields but the compounds extracted were less active, and the TEAC value was only favored at 5% ethanol, whereas at 15% ethanol the activity was reduced by half (Fig. 4b). The use of ethanol as modifier caused a similar increase on the extraction yield from *Agaricus brasiliensis* (Mazzutti et al., 2012) and from *Lentinula edodes* (Kitzberger et al., 2007).

The previous hydration-extraction of samples showed benefits in the SFE yields and, more markedly on the antiradical properties.

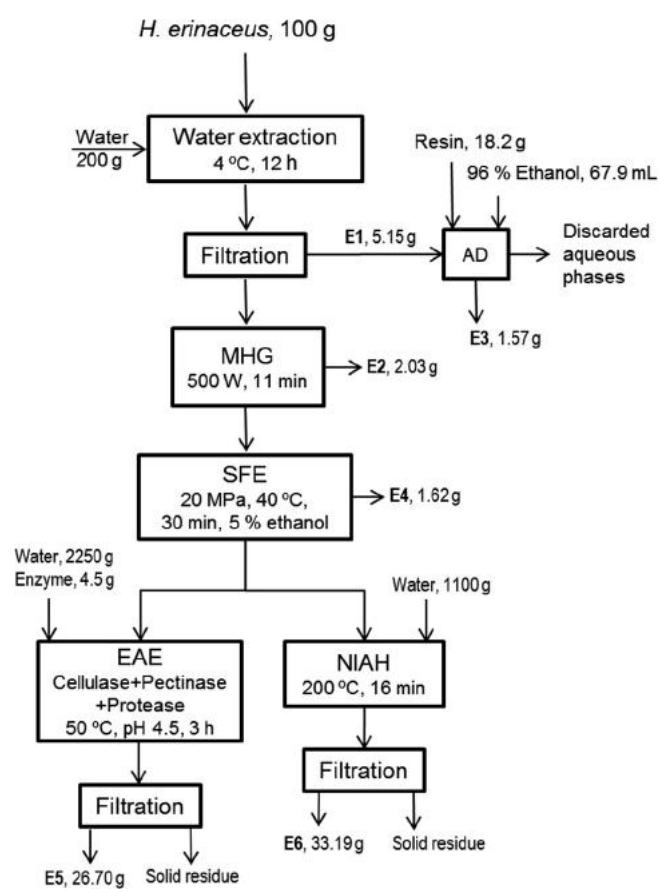

Fig. 6. Flow diagram of the sequence of stages using green solvents for the extraction of phenolic components from *Hericium erinaceus* and phenolics and saccharide yields during further enzyme aided or subcritical water extraction of the exhausted solids. AD: adsorption–desorption; MHG: microwave hydrogravity; SFE: supercritical fluid extraction; EAE: enzyme aided extraction: NIAH: non-isothermal autohydrolysis.

Table 1
Extraction yield, phenolic content and properties of the extracts obtained in different treatments of *Hericium erinaceus*.

Extraction technology (extract)		
WE (E1)		
Extraction yield (g E1/100 g *H.e.*)	5.15 ± 0.69	
Phenolic content (g GAE/100 g E1)	1.96 ± 0.11	
TEAC (g Trolox/g E1)	0.04 ± 0.01	
WE-AD (E2)		
Recovery yield (g E2/100 g *H.e.*)	1.57 ± 0.07	
Phenolic content (g GAE/100 g E2)	13.30 ± 1.05	
TEAC (g Trolox/g E2)	0.59 ± 0.05	
	Irradiation power in MHG	
	100 W	**500 W**
WE-MHG (E3)		
Extraction yield (g E3/100 g *H.e.*)	4.50 ± 0.00	4.22 ± 0.00
Phenolic content (g GAE/100 g E3)	5.16 ± 0.04	0.08 ± 0.01
TEAC (g Trolox/g E3)	0.13 ± 0.00	0.24 ± 0.02
WE-MHG-SCF (E4)		
Extraction yield (g E4/100 g *H.e.*)	0.76 ± 0.01	1.62 ± 0.00
Phenolic content (g GAE/100 g E4)	1.71 ± 0.04	3.44 ± 0.02
TEAC (g Trolox/g E4)	0.07 ± 0.02	0.22 ± 0.01
WE-MHG-SCF-EAE (E5)		
Extraction yield (g E5/100 g *H.e.*)	26.70 ± 0.38	32.36 ± 1.74
Phenolic content (g GAE/100 g E5)	1.44 ± 0.12	1.30 ± 0.09
Protein content (g BSA/100 g E5)	11.31 ± 0.24	8.17 ± 0.14
Monomers (g/100 g E5, GA; G;X + Gal + Man; FA; AA)	5.55; 0.04; 0.08; 0.28; 0.08	5.29; 0.05; 0.03; 0.48;-
Oligomers (g/100 g E5, GA; G;X; Gal; Man; Ar)	12.04; 1.16; 0.88;-;0.19; 0.22	11.21; 1.27; 0.02; 0.26; −; −
TEAC (g Trolox/g E5)	0.02 ± 0.00	0.02 ± 0.00
FRAP (mg AA/g E5)	2.73 ± 0.18	2.45 ± 0.36
WE-MHG-SCF-NIAH (E6)		
Extraction yield (g E6/100 g H.e.)	34.25 ± 4.85	33.19 ± 2.19
Phenolic content (g GAE/100 g E6)	3.33 ± 0.05	3.42 ± 0.02
Protein content (g BSA/100 g E6)	21.28 ± 0.39	17.73 ± 1.53
Monomers (g/100 g E6, GA; G;X + Gal + Man; FA; AA)	0.14; 0.00; 0.07; 0.05; 0.02	0.11; 0.01; 0.03; 0.06; 0.15
Oligomers (g/100 g E6, GA; G; X; Gal; Man; Ar)	1.73; 9.36; 0.53: 0.75; 1.33; 2.82	1.70; 10.99; 0.42; 0.73; 1.16; 4.67
TEAC (g Trolox/g E6)	0.05 ± 0.00	0.05 ± 0.00
FRAP (mg AA/g E6)	9.57 ± 0.37	9.68 ± 0.28

WE: water extraction; AD: adsorption–desorption; MHG: microwave hydrogravity; SCF: ethanol modified supercritical CO_2; EAE: enzyme assisted extraction, NIAH: non-isothermal autohydrolysis.
GA: glucuronic acid; G: glucose; X: xylose; Gal: galactose; Man: mannose; Ar: arabinose; FA: formic acid; AA: acetic acid.

The influence of the irradiation power during MHG of water hydrated-extracted samples was assessed on the supercritical CO_2 extraction performance at 20 MPa and 45 °C during 30 min with pure and with ethanol modified CO_2. Enhanced yields were attained after irradiation at 500–700 W, with a concomitant increase in the ABTS radical cation scavenging capacity. The use of cosolvent selectively improved the extraction of active components, the yields were increased by 20%, and the ABTS radical cation scavenging could be doubled (Fig. 5). The cosolvent also favored the extraction yields and radical scavenging activity of extracts from non-hydrated mushroom samples (data not shown).

3.3. Water-based fractionation

The application of further extraction stages allowed the fractionation of the raw material and the selective separation of components. The flow diagram in Fig. 6 shows the sequential extraction with green technologies and the mass balances to each stage. Alternatively, two aqueous based extraction stages have been tried to process the exhausted solids from sc-CO_2 extraction. The previous organic solvent extraction facilitates the separation of polysaccharides (Ruthes, Smiderle, & Iacomini, 2015). Digestion with a multienzyme mixture and autohydrolysis, using pressurized hot water can be further proposed to solubilize the polysaccharide fraction. Table 1 summarizes the yield and characteristics of the extracts obtained in the different processing stages. No significant influence of the irradiation power during MHG on the solubles and phenolics yields was observed, and only some difference in the radical scavenging properties of the supercritical extracts occurred.

The enzyme assisted extraction of mushroom samples previously treated in MHG at 100 W and at 500 W, yielded 26.70 ± 0.38% and 32.36 ± 1.74%, respectively. Non-isothermal autohydrolysis was performed at 200 °C, leading to maximal solubilization yields and phenolic content of the extracts; optimal temperature was in the range of that reported for the extraction of *Inonotus obliquus* under subcritical conditions using a different device and combination of time and temperature (Seo & Lee, 2010). The enzyme-assisted extraction of *H. erinaceus* in shorter incubation times was reported and the polysaccharides were mainly composed of mannose, glucose, xylose, and galactose (Zhu et al., 2014).

The non-isothermal autohydrolysis of the solids from MHG performed at 100 W and at 500 W, further extracted with supercritical carbon dioxide, yielded 34.25 ± 4.85% and 33.19 ± 2.19%, for samples irradiated at 100 W and at 500 W, respectively. The benefits of autohydrolysis and heat treatments on the phenolic content and antioxidant properties of mushrooms extracts could be related

Table 2
Mineral composition of *Hericium erinaceus* initial freeze dried samples (H. e.) and the solids remaining after (WE and MHGa t 100 or at 500 W) and the processing stages shown in the flow diagram of Fig. 6.

	Mg (mg/kg)	Ca (mg/kg)	Zn (mg/kg)	Cu (mg/kg)	Mn (mg/kg)	Cd (mg/kg)	Na (mg/kg)	K (g/kg)
H. e.	881.00 ± 29.16	333.93 ± 3.36	64.94 ± 0.57	10.55 ± 1.87	8.88 ± 0.16	<1.5	93.85 ± 4.81	31.57 ± 2.25
100 W	909.26 ± 39.14	354.11 ± 55.32	131.63 ± 0.35	15.62 ± 0.01	12.02 ± 0.33	1.91 ± 0.02	53.75 ± 5.28	18.13 ± 1.01
500 W	889.27 ± 16.87	277.85 ± 45.07	152.38 ± 5.20	16.71 ± 0.26	13.94 ± 0.40	1.96 ± 0.08	76.31 ± 17.23	15.48 ± 0.11
100 W-SCF-EAE	135.35 ± 3.31	225.67 ± 13.74	136.41 ± 4.36	19.54 ± 0.97	2.25 ± 0.27	1.80 ± 0.03	1897.75 ± 297.33	17.89 ± 0.22
500 W-SCF-EAE	213.59 ± 3.30	177.15 ± 20.34	160.77 ± 1.20	19.99 ± 0.35	4.28 ± 0.03	2.03 ± 0.02	1507.36 ± 17.27	13.35 ± 2.40
100 W-SCF-NIAH	575.04 ± 22.15	281.25 ± 62.62	188.36 ± 20.98	26.46 ± 13.37	14.77 ± 1.71	3.32 ± 0.28	104.41 ± 3.06	4.11 ± 4.85
500 W-SCF-NIAH	785.95 ± 27.55	207.27 ± 34.19	207.06 ± 4.19	28.82 ± 5.74	17.52 ± 0.91	3.33 ± 0.04	43.26 ± 32.58	2.30 ± 2.72

to the enhanced solubilization of phenolic compounds and to the formation of Maillard's reaction products, with antioxidant properties (Wong et al., 2009).

The liquid fraction also contained 0.1% mannitol and 1% arabitol. A methanol extract of the fruiting bodies of *H. erinaceus* fractionated with silica gel columns and eluted with chloroform/ethyl acetate/acetone/methanol yielded D-threitol, D-arabinitol and palmitic acid (Wang et al., 2005). The glucose oligomers were significantly more abundant in the extracts from non-isothermal autohydrolysis, accounting for 9–11% (wt). Therefore the application of this latter stage could allow a more effective fractionation of the mushroom, providing a glucan rich soluble fraction. The liquid phase from autohydrolysis contained glucose oligomers and other oligomeric components composed of xylose, galactose, rhamnose and glucuronic acid, monosaccharides (glucose, xylose, galactose and rhamnose). *H. erinaceus* polysacccharides, composed of xylose (7.8%), ribose (2.7%), glucose (68.4%), arabinose (11.3%), galactose (2.5%) and mannose (5.2%), can significantly decrease lipid peroxidation level and increase antioxidant enzymes activities in experimental mice at a dose of 300 mg/kg for 15 days (Han et al., 2013).

Both the phenolic and saccharidic fractions are the most abundant and also the ROS-generating activity and the antioxidant activity of hot water extracts from culinary-medicinal mushrooms was strongly correlated with their polysaccharide and polyphenol contents (Wei & Van Griensven, 2008), but mushrooms contain a variety of compounds such as phenolics, tocopherols, ascorbic acid, and carotenoids contributing to the antioxidant properties (Lu, Tian, Wei, & Gao, 2014) and with nutritional (Özyürek et al., 2014) and pharmacological interest (Khan et al., 2013; Kim et al., 2011).

The mineral content of the solid residues and products is summarized in Table 2. The Cr levels were under 9 mg/kg; Cd limits for other mushrooms are regulated (EU, 2014), with limits at 1.0 mg/kg fw (10 mg/kg dw) (Giannaccini et al., 2012), and the values obtained for the processed solids of this work are under these limits.

The importance of processing and not the cultivation conditions of *H. erinaceus* mycelium on the antioxidant properties of the extracts has been highlighted for the production of standardized formulations for either human nutrition or therapy (Wong et al., 2009) and the development of green processes could be suited for the integral utilization of this raw material and the production of components for nutraceutical, food and cosmetic industries.

4. Conclusions

The sequential fractionation of *H. erinaceus* components can be accomplished through a sequence of extraction stages with green solvents. Direct processing by MHG allows drying and structural alterations that facilitates supercritical CO_2 extraction. The remaining solid phase can be further hydrolyzed with either enzymes or with pressurized hot water (autohydrolysis) to solubilize the saccharidic fraction. Autohydrolysis is suited for the extraction of oligomeric glucose-rich fractions, which could be valuable for their biological properties. Further chemical characterization of these components is ongoing.

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